Tugas : Laporan Praktikum Mandiri 12

Muhammad Arifin Ilham - 0110222133

[1] Teknik Informatika, STT Terpadu Nurul Fikri, Depok
*E-mail: muha22133ti@student.nurulfikri.ac.id

```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```
path = "/content/gdrive/MyDrive/machine learning/praktikum/praktikum12"
```

Import Library

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.decomposition import PCA
from sklearn.metrics import accuracy_score, classification_report
```

Load Data

```
df = pd.read_csv(path + '/data/data.csv', sep=",")

df = df.drop(['id', 'Unnamed: 32'], axis=1)

df['diagnosis'] = df['diagnosis'].map({'M': 1, 'B': 0})

X = df.drop('diagnosis', axis=1)
y = df['diagnosis']

print("Shape X:", X.shape)
print("Shape y:", y.shape)
df.head()
```

```
Shape X: (569, 30)
Shape y: (569,)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
0	1	17.99	10.38	122.80	1001.0	0.11840
1	1	20.57	17.77	132.90	1326.0	0.08474
2	1	19.69	21.25	130.00	1203.0	0.10960
3	1	11.42	20.38	77.58	386.1	0.14250
4	1	20.29	14.34	135.10	1297.0	0.10030

5 rows × 31 columns

Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print("Data train:", X_train.shape)
print("Data test:", X_test.shape)
```

```
Data train: (455, 30)
Data test: (114, 30)
```

Standarisasi Data

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Penerapan Model SVM Tanpa PCA

```
svm_no_pca = SVC(kernel='rbf', random_state=42)
svm_no_pca.fit(X_train_scaled, y_train)

y_pred_no_pca = svm_no_pca.predict(X_test_scaled)

acc_no_pca = accuracy_score(y_test, y_pred_no_pca)
print("Akurasi SVM tanpa PCA:", acc_no_pca)
```

```
Akurasi SVM tanpa PCA: 0.9736842105263158
```

Penerapan PCA

```python
pca = PCA()
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
```

Variansi yang Dijelaskan oleh Setiap Komponen PCA

```python
explained_variance = pca.explained_variance_ratio_

for i, var in enumerate(explained_variance):
    print(f"PC{i+1}: {var:.4f}")
```

```
PC1: 0.4459
PC2: 0.1855
PC3: 0.0958
PC4: 0.0659
PC5: 0.0562
PC6: 0.0399
PC7: 0.0221
PC8: 0.0161
PC9: 0.0128
PC10: 0.0117
PC11: 0.0101
PC12: 0.0091
PC13: 0.0079
PC14: 0.0048
PC15: 0.0030
PC16: 0.0025
PC17: 0.0020
PC18: 0.0017
PC19: 0.0016
PC20: 0.0011
PC21: 0.0010
PC22: 0.0008
PC23: 0.0008
PC24: 0.0005
PC25: 0.0005
PC26: 0.0003
PC27: 0.0002
PC28: 0.0001
PC29: 0.0000
PC30: 0.0000
```

Membangun Model SVM dengan PCA (3 Komponen)

```python
pca_3 = PCA(n_components=3)
X_train_pca3 = pca_3.fit_transform(X_train_scaled)
X_test_pca3 = pca_3.transform(X_test_scaled)

svm_pca = SVC(kernel='rbf', random_state=42)
svm_pca.fit(X_train_pca3, y_train)

y_pred_pca = svm_pca.predict(X_test_pca3)
acc_pca = accuracy_score(y_test, y_pred_pca)

print("Akurasi SVM dengan PCA:", acc_pca)
```

```
Akurasi SVM dengan PCA: 0.956140350877193
```

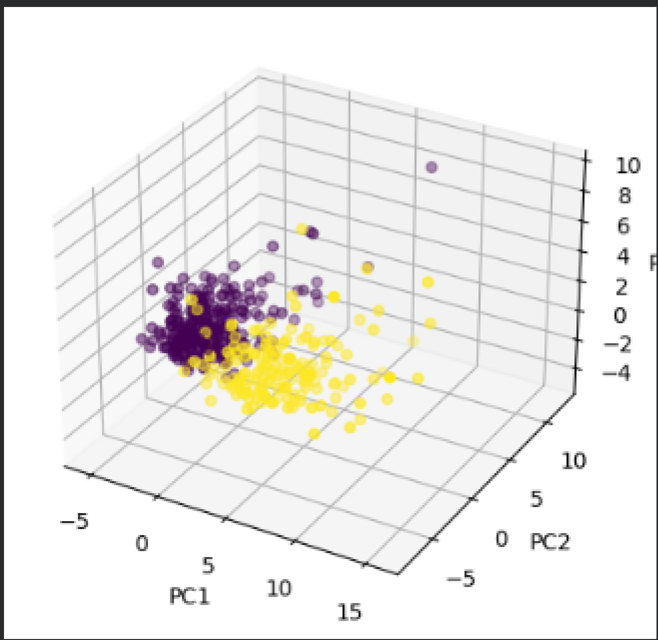
Visualisasi PCA dalam Ruang 3 Dimensi

```python
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

scatter = ax.scatter(
    X_train_pca3[:, 0],
    X_train_pca3[:, 1],
    X_train_pca3[:, 2],
    c=y_train
)

ax.set_xlabel("PC1")
ax.set_ylabel("PC2")
ax.set_zlabel("PC3")
plt.show()
```



Perbandingan Model SVM Tanpa PCA dan Dengan PCA

```python
comparison = pd.DataFrame({
    "Model": ["SVM tanpa PCA", "SVM dengan PCA"],
    "Akurasi": [acc_no_pca, acc_pca]
})

comparison
```

	Model	Akurasi
0	SVM tanpa PCA	0.973684
1	SVM dengan PCA	0.956140

Link GitHub: https://github.com/CeperMail/MachineLearning.git